

Requirement specification

Project Title	VoteVox
Project Leader	Hubauer Paul Matthias
Created on	13.11.2025
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State	Completed
Current Version	1.3

Change History

Nr.	Date	Version	Changed Chapter	Type of Change	Author	State
1	13.11.2025	1.1	1, 2, 3, 4	Create Content	Aigner Sandro	started
2	27.11.2025	1.2	5, 6	Create Content	Aigner Sandro	in progress
3	08.12.2025	1.3	7,8	Create Content	Hubauer Paul Matthias	Completed

1. Introduction

The present requirements specification defines the functional and non-functional requirements for the VoteVox application. The document serves as the basis for tendering, development, and later contractual agreements between the client and the contractor. Once a contract is concluded, this requirements' specification becomes legally binding. Any prior agreement loses their validity unless explicitly stated otherwise in this document.

VoteVox is intended to support schools in organizing democratic decision processes such as class representative elections, survey-based decisions, or anonymous voting.

2. General

❖ Aim and purpose of the document

This document describes the specific requirements, functions, and goals of the VoteVox project. It provides an overview of all functional and technical requirements as well as conditions that must be met to complete the project.

❖ Initial situation

Currently, you receive a QR code to vote,
but you cannot see how many votes each candidate received.

❖ Abbreviations

- UI – User Interface
- API – Application Programming Interface
- OTP – One-Time Password
- ID – Identification Number
- GDPR - General Data Protection Regulation

❖ Teams and Interfaces

Role	Name	Telephone	E-Mail	Team
Project Manager	Hubauer Paul Matthias	+43 123 4560001	paul.hubauer@htl-wels.at	AB001
Project Team Member	Aigner Sandro	+43 123 4560002	sandro.aigner@htl-wels.at	AB002
Project Team Member	Anzengruber Laurenz	+43 123 4560003	laurenz.anzengruber@htl-wels.at	AB003

3. Concept

- ❖ Goal(s) of the provider specification
 - Create a secure, easy-to-use voting app for schools.
 - Make voting transparent, anonymous and fast.
 - Provide an administration dashboard for teachers.
- ❖ Specifications Goals and benefits for the user
 - Students can participate in elections easily and anonymously.
 - Teachers can set up elections within minutes.
 - Administrators can monitor voting participation and results.
 - Results are calculated automatically and accurately.
- ❖ Target group(s)
 - Students
 - Teachers

4. Functional Requirements

- ❖ User Registration
 - Students log in using a school-generated code or ID.
- ❖ Election Creation
 - Teachers can create elections with:
 - Title
 - Description
 - Start and end time
 - Eligible participants
- ❖ Voting Process
 - Each student can cast exactly one vote per election.
 - Votes are stored securely and cannot be altered afterwards.
- ❖ Result Overview
 - Automatic result calculation.
 - Results can be displayed as:
 - Bar charts
 - Percentages

5. Non-Functional Requirements

❖ General Requirements

- The app must be intuitive and easy to use
- The system must support at least five hundred simultaneous users

❖ Legal Requirements

- Comply with GDPR
- No unnecessary data must be collected.

❖ Technical Requirements

- The system must work on web browsers.

6. Framework conditions

❖ Schedule

Task	Who	When
Analyse Competition	Project Team Members	Week 1
Research Regulation	Project Manager	Week 1
Actual Development	Full Team	Week 2-5
Testing	Full Team	Week 6

❖ Technical Requirements

Hardware Requirements:

Minimum: PC	Recommended: PC
Dual-core CPU (2.0 GHz or higher)	Quad-core CPU (3.0 GHz or higher)
4 GB RAM	8 GB RAM
256 GB storage (HDD or SSD)	512 GB SSD
Stable local network connection	
Minimum: Phone	Recommended: Phone
Android 8+ or iOS 13+ device	Android 11+ or iOS 15+
Rear camera with at least 5 MP	8–12 MP camera for faster QR code focus
1 GB RAM	2–4 GB RAM
Stable internet connection	Stable Internet connection
Browser with camera access support (Chrome, Safari, Firefox)	

Software Recommended Requirements:

Software Requirements (Admin/Server PC)

- Windows 10+ / macOS 12+ / Linux Ubuntu 22+
- Node.js or Python (depending on implementation)
- Database software: PostgreSQL / MySQL / SQLite
- Modern browser (Chrome, Firefox, Edge)
- PDF viewer (for checking exported QR codes)
- QR code generation libraries (e.g., qrcode.js, Python qrcode.py)
- Printer drivers for QR code sheet printing

Software Requirements (Student Smartphones)

- Browser with WebRTC or getUserMedia support for live camera scanning
- QR scanning permissions enabled
- JavaScript enabled
- No additional apps required (fully browser-based)

❖ Problem Analysis

Problem	Solution
Traditional voting processes are slow, error-prone, and difficult to keep anonymous.	Digital, automated voting system with unique QR codes ensuring anonymity and accurate counting.
Students might attempt to vote multiple times or participate in elections they are not eligible for.	Single-use QR codes with token validation and election-type authorization checks.
Manual counting and evaluation of ballots requires significant time and effort.	Automated data storage, real-time statistics, and export functions (PDF, Excel/CSV).
Strict data protection requirements (e.g., GDPR) must be met.	Separation of identity and vote data, anonymized storage, and restricted access through an admin panel.

7. Delivery and Acceptance conditions

- ❖ Version schedule (planned)

Development Finish: end April 2026
Start of Alpha Testing: early May 2026
Alpha Version Release: 13 May 2026

- ❖ Pricing

One-Time-Fee per Election (0,10€/Student) + Yearly Maintenance Fee (199€)

8. Appendix

Project Handbook



VoteVox_Handbook.pdf

UML Diagrams



VoteVox_Class_Diagram.pdf



VoteVox_Component_Diagram.pdf



VoteVox_Sequence_Diagram_Management



VoteVox_Sequence_Diagram_Student



VoteVox_Use_Case_Diagram